

The finite element method

The Dirichlet problem

Given boundary values g , find a harmonic function u such that

$$\Delta u = 0 \text{ in } \Omega, \quad u = g \text{ on } \partial\Omega.$$

Last lecture showed that such a function is unique. Today we construct it using finite elements.

In practice: how does one compute the solution?

Uniqueness does not tell us what the solution is, or that there is one. To find it we use the **finite element method**. Three steps:

1. rewrite the problem in **weak form**;
2. **choose** where to look: place a grid on Ω , replace u by a function built from finitely many pieces, and $u = g$ by a condition at finitely many boundary points;
3. **solve** the linear system that results.

By the end we will have a finite-dimensional picture of why the boundary values determine everything, and a sequence of linear-algebra solutions converging to a function satisfying the weak form, under the grid assumptions stated below.

Step 1: the weak form

Take any function w vanishing on $\partial\Omega$. Multiply $\Delta u = 0$ by w , integrate, and integrate by parts (divergence theorem for $w \nabla u$; the boundary term vanishes because $w = 0$ there):

$$\begin{aligned} 0 &= \iint_{\Omega} w \Delta u \, dA = \oint_{\partial\Omega} w \nabla u \cdot n \, ds - \iint_{\Omega} \nabla w \cdot \nabla u \, dA \\ &= - \iint_{\Omega} \nabla w \cdot \nabla u \, dA. \end{aligned}$$

Weak form

Write $a(u, w) := \iint_{\Omega} \nabla u \cdot \nabla w \, dA$. Find u with $u = g$ on $\partial\Omega$ such that $a(u, w) = 0$ for every w vanishing on $\partial\Omega$.

Step 2: a grid and hat functions

Assume Ω is a union of little squares of side ε , with corners (the **nodes**) at grid points (x_i, y_j) .

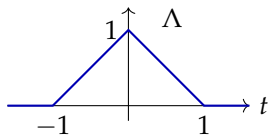
Tent function in one variable:

$$\Lambda(t) := \max(0, 1 - |t|).$$

Basis function at a node:

$$\phi_{ij}(x, y) := \Lambda\left(\frac{x - x_i}{\varepsilon}\right) \Lambda\left(\frac{y - y_j}{\varepsilon}\right).$$

A tent of height 1 over (x_i, y_j) . Its classical gradient exists away from the grid lines and is square-integrable; the grid lines have zero area.



Step 2: nodal values

Key property. $\phi_{ij}(x_k, y_l) = 1$ if $(k, l) = (i, j)$, and 0 at every other node.

Hence a linear combination

$$U = \sum_{ij} c_{ij} \phi_{ij} \quad \text{satisfies} \quad U(x_i, y_j) = c_{ij} :$$

the coefficients are the nodal values.

These U are exactly the continuous functions that are **bilinear** on each square, of the form $a + bx + cy + dxy$ there: linear along every grid line, but not linear on the square, because of the xy term. Call their space V_ε ; it is the span of the ϕ_{ij} .

Step 2: the spaces V_ε^g and V_ε^0

A node is **interior** if all four squares around it lie in Ω , and a **boundary node** otherwise. Two subsets of V_ε matter:

- ▶ V_ε^g : the $U \in V_\varepsilon$ with $U = g$ at every boundary node. Not a linear space (the sum of two such functions is $2g$ there), but an affine one: averages of its members stay in it.
- ▶ V_ε^0 : the $U \in V_\varepsilon$ that vanish at every boundary node, hence on all of $\partial\Omega$ (bilinear is linear along an edge). A linear space, spanned by the hats at interior nodes.

We look for the solution in V_ε^g : **finitely many unknowns**, one per interior node.

Step 3: the linear system

Ansatz. $U = \sum_{ij} c_{ij} \phi_{ij}$, with $c_{ij} = g(x_i, y_j)$ at the boundary nodes (known: the second choice of Step 2) and c_{ij} unknown at the interior nodes.

Test against each interior ϕ_{kl} . It vanishes on $\partial\Omega$, so it is an admissible w . Put $u = U, w = \phi_{kl}$ in the weak form:

$$\sum_{ij} A_{kl,ij} c_{ij} = 0 \quad \text{for every interior node } (k,l),$$
$$\text{where } A_{kl,ij} := a(\phi_{ij}, \phi_{kl}) = \iint_{\Omega} \nabla \phi_{ij} \cdot \nabla \phi_{kl} dA.$$

One equation per interior node, one unknown per interior node: an $n \times n$ system, whatever the shape of Ω .

Step 3: block form

Let I be the interior nodes and B the boundary nodes, and split the coefficient vector accordingly: c_I unknown, $c_B = g_B := (g(x_i, y_j))_{(i,j) \in B}$ known.

Split the matrix the same way,

$$A_{II} = (A_{pq})_{p,q \in I}, \quad A_{IB} = (A_{pq})_{p \in I, q \in B}.$$

Then the equations read

$$A_{II} c_I = -A_{IB} g_B.$$

The boundary data is on the right, the unknowns on the left. What remains is to see that A_{II} is invertible.

Step 3: the system has exactly one solution

A is symmetric, since $a(\phi_{ij}, \phi_{kl}) = a(\phi_{kl}, \phi_{ij})$. Its quadratic form is the energy: for any coefficients c and the corresponding $U = \sum c_{ij} \phi_{ij}$,

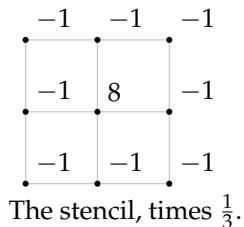
$$c^T A c = \sum_{ij, kl} c_{kl} A_{kl, ij} c_{ij} = a(U, U) = \iint_{\Omega} |\nabla U|^2 dA \geq 0.$$

Now let c be supported on the interior nodes, so $U \in V_{\varepsilon}^0$. If $c^T A c = 0$ then $\nabla U \equiv 0$, so U is constant on each connected component of Ω . Each component has boundary points, where $U = 0$. So $U \equiv 0$ and $c = 0$.

So A_{II} is **positive definite**, hence invertible, and the system has exactly one solution c_I .

What the equations say

$A_{kl,ij} \neq 0$ only when the nodes (i,j) and (k,l) share a square, i.e. are neighbours. Integrating over one square and summing over the squares that meet (k,l) gives the stencil on the right: $-\frac{1}{6}$ to each edge neighbour from each of the two squares they share, $-\frac{1}{3}$ to each diagonal neighbour from the one square they share, and $\frac{8}{3}$ on the diagonal.



So the equation at an interior node is

$$c_{kl} = \frac{1}{8} \sum_{8 \text{ neighbours } (i,j) \text{ of } (k,l)} c_{ij}.$$

Each interior value is the average of its eight neighbours.

The discrete maximum principle

If an interior node has the global maximum value, its average equation forces all eight neighbours to have that same value. Repeating this argument propagates the value to the boundary; the same reasoning applies to a minimum. Moreover, a bilinear function on a square attains its extrema at the corners. Thus U_ε attains its maximum and its minimum at boundary nodes. This is the **discrete maximum principle**, and it has two consequences.

The discrete maximum principle: consequences

- ▶ **Uniqueness, again.** With zero boundary data both extremes are 0, so $U_\varepsilon \equiv 0$.
- ▶ **A bound.** $\min_{\partial\Omega} g \leq U_\varepsilon \leq \max_{\partial\Omega} g$, uniformly in ε .

So the solution is a weighted average of the boundary data, and the linear system computes the weights. That is what “harmonic functions are determined by their boundary values” means concretely.

Existence: the discrete solution minimizes the energy

Up to now we have assumed that a solution exists. The discrete problem will produce one.

Lemma

Among all $V \in V_\varepsilon^g$, U_ε is the unique minimizer of the **energy**
 $E(V) := a(V, V) = \iint_\Omega |\nabla V|^2 dA$.

For any $V \in V_\varepsilon^g$, write $V = U_\varepsilon + w$ with $w \in V_\varepsilon^0$. Then

$$\begin{aligned} E(V) &= E(U_\varepsilon) + 2a(U_\varepsilon, w) + E(w) \\ &= E(U_\varepsilon) + E(w) \geq E(U_\varepsilon), \end{aligned}$$

because the discrete equations give $a(U_\varepsilon, w) = 0$. Equality forces $E(w) = 0$, hence $w = 0$ by positive definiteness on V_ε^0 . Thus U_ε is the unique minimizer.

□

Existence: the minimizers are Cauchy

Refine dyadically, $\varepsilon_n := 2^{-n}\varepsilon_0$. Assume for now that g is linear on each boundary edge of the coarsest grid. Then “equals g at the boundary nodes” means the same thing on every grid, the sets $V_{\varepsilon_n}^g$ are nested, and the minimum $m_n := E(U_{\varepsilon_n})$ over a growing set decreases. It is ≥ 0 , so $m_n \downarrow m$.

Existence: the parallelogram law

The symmetric bilinear form a satisfies the parallelogram identity $E(x - y) + E(x + y) = 2E(x) + 2E(y)$: expand both sides and cancel the cross terms. For a scalar- or vector-valued function F , write $\|F\|_{L^2}^2 := \iint_{\Omega} |F|^2 dA$. Taking $x = U_n$ and $y = U_k$ gives

$$\begin{aligned}\|\nabla(U_n - U_k)\|_{L^2}^2 &= 2E(U_n) + 2E(U_k) - E(U_n + U_k) \\ &= 2m_n + 2m_k - 4E\left(\frac{1}{2}(U_n + U_k)\right),\end{aligned}$$

using $E(U_n) = m_n$ (U_n is the minimizer at level n) and $E(U_n + U_k) = 4E\left(\frac{1}{2}(U_n + U_k)\right)$ (the energy is quadratic).

Existence: the bound tends to zero

For $n \leq k$ both U_n and U_k lie in $V_{\varepsilon_k}^g$, and so does their average (the boundary values average to g), so its energy is at least the minimum over $V_{\varepsilon_k}^g$:

$E(\frac{1}{2}(U_n + U_k)) \geq m_k \geq m$. Hence

$$\begin{aligned}\|\nabla(U_n - U_k)\|_{L^2}^2 &\leq 2m_n + 2m_k - 4m \\ &= 2(m_n - m) + 2(m_k - m) \longrightarrow 0 \quad (n, k \rightarrow \infty),\end{aligned}$$

because $m_n \downarrow m$.

Existence: Poincaré's inequality

Poincaré's inequality

If v is continuous, piecewise smooth, and $v = 0$ on $\partial\Omega$, then

$\|v\|_{L^2} \leq L \|\nabla v\|_{L^2}$, where L is the width of Ω .

Proof. Extend v by zero outside Ω ; it stays continuous and piecewise smooth. Place Ω in the strip $0 < x < L$. For each y , $v(x, y) = \int_0^x \partial_x v(s, y) ds$, so by Cauchy–Schwarz

$$|v(x, y)|^2 \leq x \int_0^L |\partial_x v(s, y)|^2 ds \leq L \int_0^L |\partial_x v(s, y)|^2 ds.$$

Existence: Poincaré's inequality, concluded

Integrate over $x \in (0, L)$ and over y :

$$\iint |v|^2 \leq L^2 \iint |\partial_x v|^2 \leq L^2 \iint |\nabla v|^2. \quad \square$$

It applies to $U_n - U_k$. This function lies in $V_{\varepsilon_k}^0$: it is continuous, bilinear on each square, and zero at every boundary node.

On a boundary edge a bilinear function is linear, and a linear function that vanishes at both endpoints vanishes on the whole edge. So $U_n - U_k = 0$ on all of $\partial\Omega$, and Poincaré's inequality applies with $C = L$.

Existence: the limit satisfies the weak form

Since $U_n - U_k \in V_{\varepsilon_k}^0$, Poincaré's inequality gives $\|U_n - U_k\|_{L^2} \leq C \|\nabla(U_n - U_k)\|_{L^2}$. Thus U_n , $\partial_x U_n$, and $\partial_y U_n$ are Cauchy in L^2 ; call their limits u, p, q . For every smooth ψ supported inside Ω , integration by parts and passage to the limit give

$$\iint u \partial_x \psi = - \iint p \psi, \quad \iint u \partial_y \psi = - \iint q \psi.$$

By definition, $p = \partial_x u$ and $q = \partial_y u$ are the **weak derivatives** of u .

Any $w \in V_{\varepsilon_N}^0$ is a test function at every level $n \geq N$, so $a(U_n, w) = 0$ passes to the limit: $a(u, w) = 0$. For any smooth zero-boundary test function η , its nodal interpolants $\eta_N \in V_{\varepsilon_N}^0$ satisfy $\|\nabla(\eta - \eta_N)\|_{L^2} \rightarrow 0$ by Taylor's theorem on each square. Passing to the limit once more gives $a(u, \eta) = 0$ for every such η .

Existence: what we have proved

Assume that Ω is a finite union of squares from the coarsest grid and that g is linear on each boundary edge of that grid. Let $U_0 \in V_{\varepsilon_0}^g$ be the coarsest solution; it and every U_n equal g on all of $\partial\Omega$.

1. $U_n \rightarrow u$ in $L^2(\Omega)$ and $\nabla U_n \rightarrow \nabla u$ in $L^2(\Omega)$: u is a square-integrable function with a square-integrable (weak) gradient.
2. $a(u, w) = 0$ for every smooth w vanishing on $\partial\Omega$.
3. The boundary condition is retained as follows: every $U_n - U_0$ vanishes on $\partial\Omega$, while $U_n - U_0 \rightarrow u - U_0$ and $\nabla(U_n - U_0) \rightarrow \nabla(u - U_0)$ in L^2 .

Thus u satisfies the **weak form** of the Dirichlet problem. Two things are missing, both next lecture:

- ▶ u is smooth and $\Delta u = 0$ pointwise: **Weyl's lemma**;
- ▶ $u(x) \rightarrow g(p)$ as $x \rightarrow p \in \partial\Omega$, pointwise: **the random walk**.